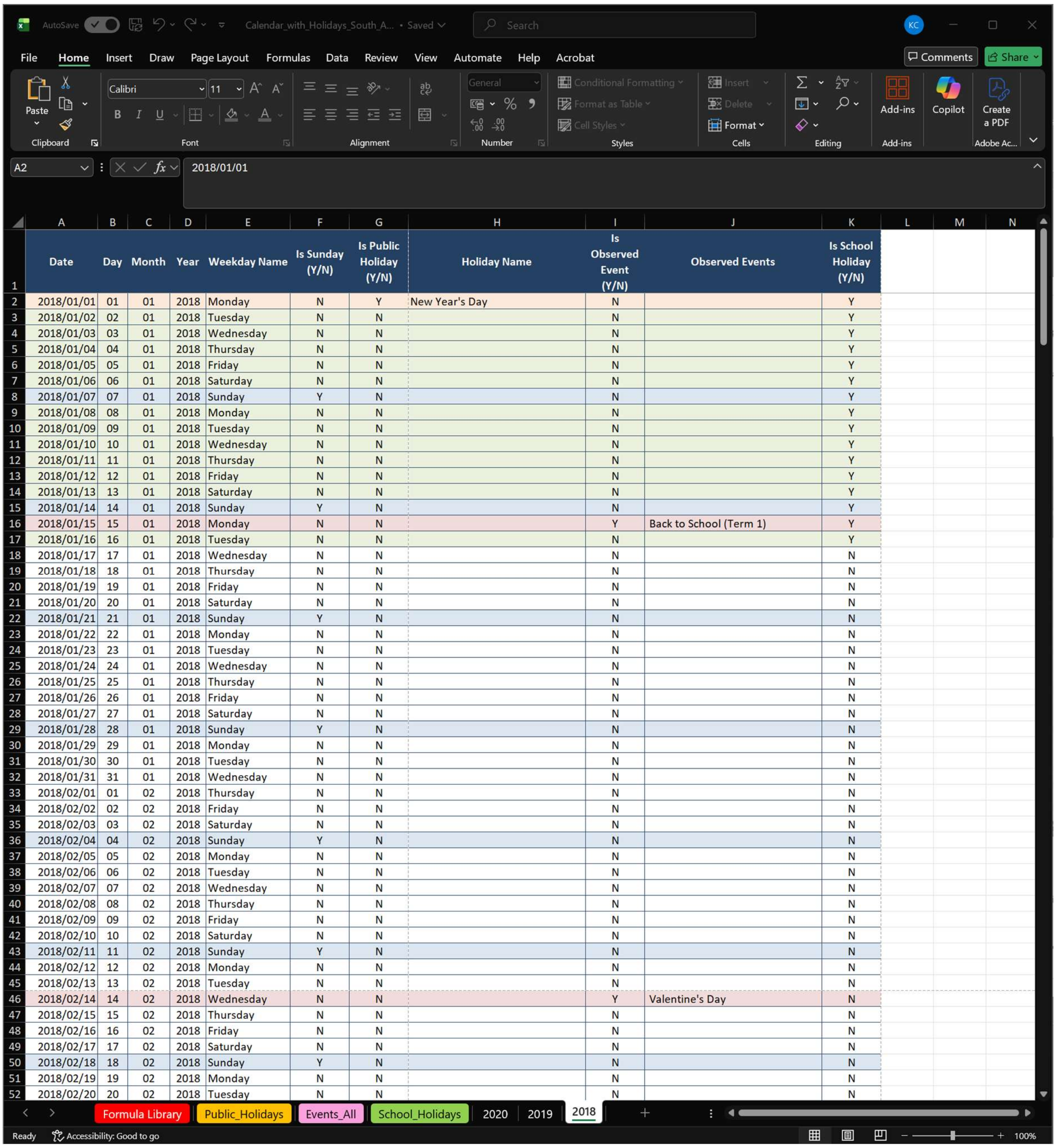
South African Retail Calendar Automation Engine (Excel)

South African Retail Calendar Automation Engine (Excel)
A modular, multi-year calendar system integrating Public Holidays, Events, and School Holidays.

Overview of Workbook:

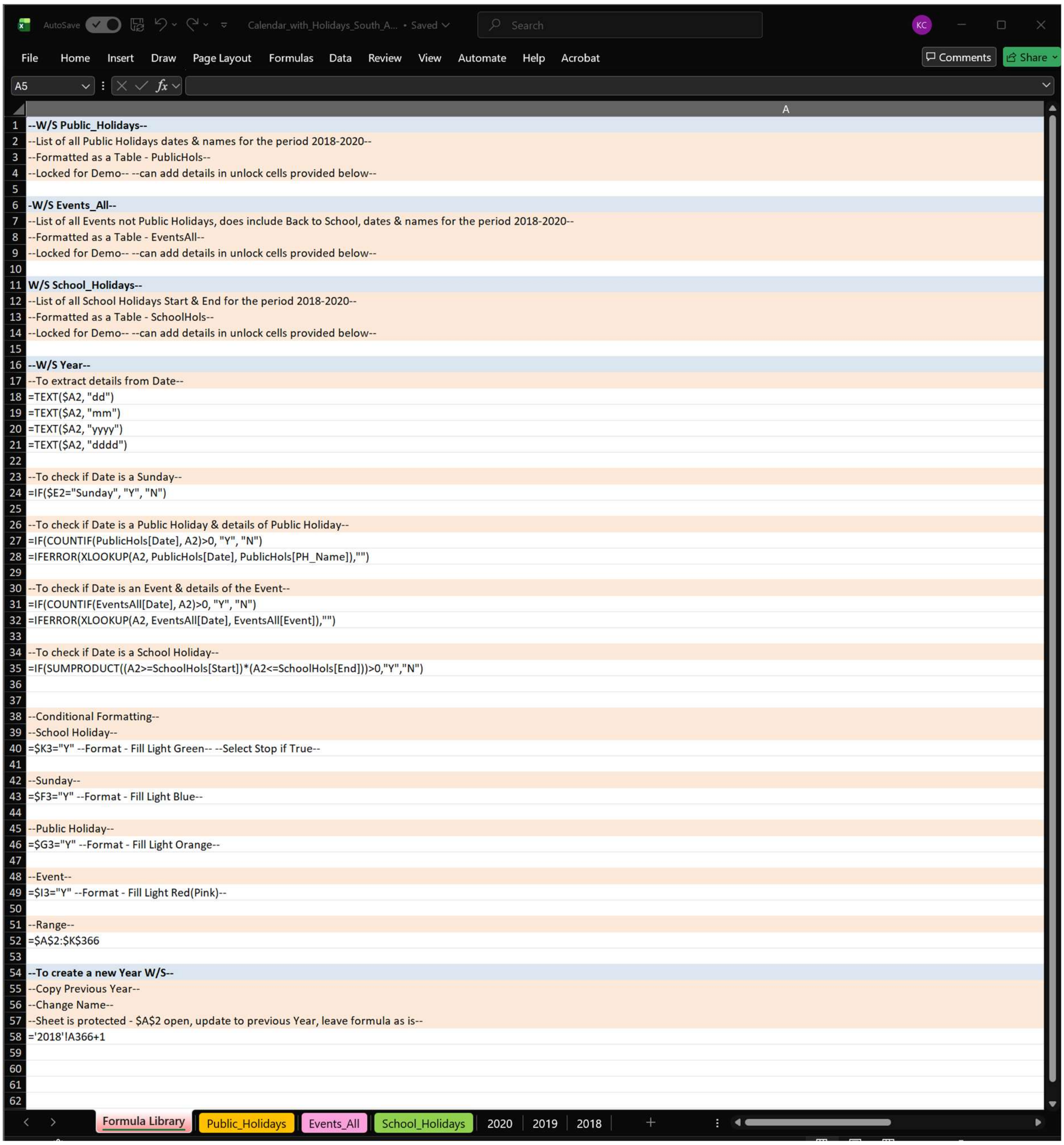


Date	Day	Month	Year	Weekday Name	Is Sunday (Y/N)	Is Public Holiday (Y/N)	Holiday Name	Is Observed Event (Y/N)	Observed Events	Is School Holiday (Y/N)
2018/01/01	01	01	2018	Monday	N	Y	New Year's Day	N		Y
2018/01/02	02	01	2018	Tuesday	N	N		N		Y
2018/01/03	03	01	2018	Wednesday	N	N		N		Y
2018/01/04	04	01	2018	Thursday	N	N		N		Y
2018/01/05	05	01	2018	Friday	N	N		N		Y
2018/01/06	06	01	2018	Saturday	N	N		N		Y
2018/01/07	07	01	2018	Sunday	Y	N		N		Y
2018/01/08	08	01	2018	Monday	N	N		N		Y
2018/01/09	09	01	2018	Tuesday	N	N		N		Y
2018/01/10	10	01	2018	Wednesday	N	N		N		Y
2018/01/11	11	01	2018	Thursday	N	N		N		Y
2018/01/12	12	01	2018	Friday	N	N		N		Y
2018/01/13	13	01	2018	Saturday	N	N		N		Y
2018/01/14	14	01	2018	Sunday	Y	N		N		Y
2018/01/15	15	01	2018	Monday	N	N		Y	Back to School (Term 1)	Y
2018/01/16	16	01	2018	Tuesday	N	N		N		Y
2018/01/17	17	01	2018	Wednesday	N	N		N		N
2018/01/18	18	01	2018	Thursday	N	N		N		N
2018/01/19	19	01	2018	Friday	N	N		N		N
2018/01/20	20	01	2018	Saturday	N	N		N		N
2018/01/21	21	01	2018	Sunday	Y	N		N		N
2018/01/22	22	01	2018	Monday	N	N		N		N
2018/01/23	23	01	2018	Tuesday	N	N		N		N
2018/01/24	24	01	2018	Wednesday	N	N		N		N
2018/01/25	25	01	2018	Thursday	N	N		N		N
2018/01/26	26	01	2018	Friday	N	N		N		N
2018/01/27	27	01	2018	Saturday	N	N		N		N
2018/01/28	28	01	2018	Sunday	Y	N		N		N
2018/01/29	29	01	2018	Monday	N	N		N		N
2018/01/30	30	01	2018	Tuesday	N	N		N		N
2018/01/31	31	01	2018	Wednesday	N	N		N		N
2018/02/01	01	02	2018	Thursday	N	N		N		N
2018/02/02	02	02	2018	Friday	N	N		N		N
2018/02/03	03	02	2018	Saturday	N	N		N		N
2018/02/04	04	02	2018	Sunday	Y	N		N		N
2018/02/05	05	02	2018	Monday	N	N		N		N
2018/02/06	06	02	2018	Tuesday	N	N		N		N
2018/02/07	07	02	2018	Wednesday	N	N		N		N
2018/02/08	08	02	2018	Thursday	N	N		N		N
2018/02/09	09	02	2018	Friday	N	N		N		N
2018/02/10	10	02	2018	Saturday	N	N		N		N
2018/02/11	11	02	2018	Sunday	Y	N		N		N
2018/02/12	12	02	2018	Monday	N	N		N		N
2018/02/13	13	02	2018	Tuesday	N	N		N		N
2018/02/14	14	02	2018	Wednesday	N	N		Y	Valentine's Day	N
2018/02/15	15	02	2018	Thursday	N	N		N		N
2018/02/16	16	02	2018	Friday	N	N		N		N
2018/02/17	17	02	2018	Saturday	N	N		N		N
2018/02/18	18	02	2018	Sunday	Y	N		N		N
2018/02/19	19	02	2018	Monday	N	N		N		N
2018/02/20	20	02	2018	Tuesday	N	N		N		N

Summary

This tool is intended for Retail Managers, Operations Analysts, Planners, and anyone responsible for forecasting labour, planning trading patterns, or coordinating seasonal campaigns. By integrating all critical dates into one automated framework, the system supports better staffing decisions, aligns labour budgets with trading intensity, and ensures promotional calendars are built on accurate, region-specific timelines. The result is a faster, more informed planning process that improves operational readiness and supports data-driven decision-making across the business.

Overview – Function Library: Intended to minimise user inputs



- **Date Breakdown**

Each date in the calendar automatically generates its own supporting details, including the day, month, year, and weekday name. This allows the system to understand the structure of the year without requiring any manual input. By breaking each date into its components, the calendar can drive downstream logic such as identifying weekends, aligning trading patterns, and supporting operational planning.

- **Holiday & Event Detection**

The system cross-checks every date against three structured tables: Public Holidays, Retail Events, and School Holidays. When a match is found, the calendar automatically flags the date and retrieves the corresponding name or event description. This ensures that all key trading days are identified consistently and accurately, without relying on manual updates or external references.

- **Colour-Coded Indicators**

To make the calendar easy to interpret at a glance, the system applies colour-coded highlights for Sundays, Public Holidays, Events, and School Holidays. These visual cues help planners quickly identify high-impact days, peak trading periods, and operational constraints. The colour logic is fully automated and updates instantly when new years are generated.

- **Observed Holidays**

When a Public Holiday falls on a weekend, the system automatically identifies the observed day (typically the following Monday). This ensures that staffing, trading hours, and operational planning reflect the actual business impact rather than just the calendar date. Observed holidays are treated with the same logic and visibility as standard holidays.

- **School Holiday Ranges**

Instead of listing every school holiday date individually, the system uses start-and-end ranges to determine whether a given date falls within a school break. This approach reduces maintenance, improves performance, and ensures accuracy across multi-week periods. School holidays are highlighted and flagged automatically, supporting workforce planning and forecasting for peak family-shopping periods.

- **Scalability**

The system is fully scalable, allowing new years, additional events, and expanded holiday datasets to be incorporated without modifying any underlying logic. Its table-driven design ensures that the calendar can grow with operational needs while maintaining accuracy, consistency, and minimal user input.

- **Protection Strategy**

The workbook is intentionally designed with a controlled structure to protect the integrity of the system while keeping user interaction simple and safe. All formulas and logic cells are locked to prevent accidental overwrites, ensuring that the underlying calculations remain consistent across years. Only the empty table rows in the lookup sheets are left open, allowing users to add new Public Holidays, Events, or School Holiday ranges without risking damage to the system.

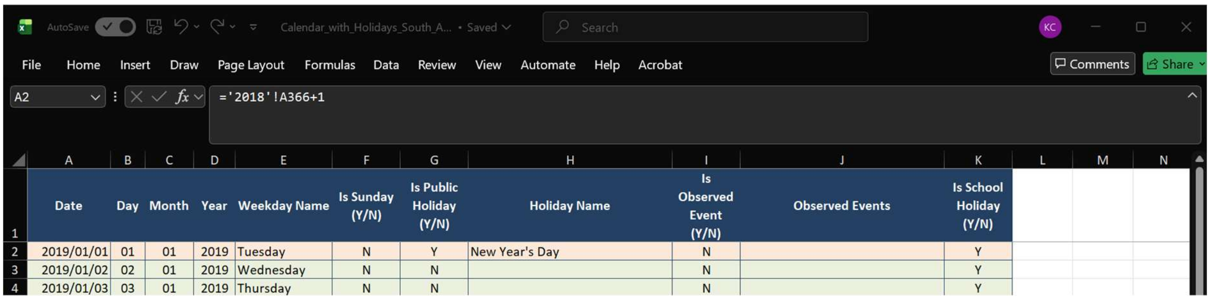
On each Year sheet, only cell **A2** is unlocked. This single input drives the entire year-generation process, meaning users can create new years without touching any formulas or layout elements. The workbook structure itself is protected to prevent renaming, deleting, or moving sheets, preserving the architecture that the system depends on.

Together, these protections create a stable, low-maintenance environment that supports long-term use. Users can safely update data and generate new years without technical knowledge, while the system remains consistent, reliable, and resistant to accidental changes.

Creating a New Year

Steps:

1. Copy previous year's worksheet
2. Rename sheet
3. Update A2, by only changing the year
4. Everything updates automatically



A2 ='2018'!A366+1

Date	Day	Month	Year	Weekday Name	Is Sunday (Y/N)	Is Public Holiday (Y/N)	Holiday Name	Is Observed Event (Y/N)	Observed Events	Is School Holiday (Y/N)
2019/01/01	01	01	2019	Tuesday	N	Y	New Year's Day	N		Y
2019/01/02	02	01	2019	Wednesday	N	N		N		Y
2019/01/03	03	01	2019	Thursday	N	N		N		Y

Skills Demonstrated

Advanced Excel
Structured Tables
Lookup Logic
Conditional Formatting

System Architecture
Data Modelling
Retail Operations Knowledge